McKinsey & Company

Transformation and Strategy & Corporate Finance Practices

How capital expenditure management can drive performance

Want to reduce project costs and timelines while increasing returns? Undertake a top-to-bottom reassessment of your capital investments at every stage of the life cycle.

by Tom Brinded, Erikhans Kok, Lucas Ponbauer, and Bevan Watson

© Daniela Duncan/Getty Images

June 2022

One of the quickest and most effective ways for organizations to preserve cash is to reexamine their capital investments. The past two years have offered a fascinating look into how different sectors have weathered the COVID-19 storm: from the necessarily capital expenditure–starved airport industry to the cresting wave of public-sector investments in renewable infrastructure and anticipation of the next mining supercycle. Indeed, companies that reduce spending on capital projects can both quickly release significant cash and increase ROIC, the most important metric of financial value creation (Exhibit 1).

This strategy is even more vital in competitive markets, where ROIC is perilously close to cost of capital. In our experience, organizations that focus on actions across the whole project life cycle, the capital project portfolio, and the necessary foundational enablers can reduce project costs and timelines by up to 30 percent to increase ROIC by 2 to 4 percent. Yet managing capital projects is complex, and many organizations struggle to extract cost savings. In addition, ill-considered cuts to key projects in a portfolio may actually jeopardize future operating performance and outcomes. This dynamic reinforces the age-old challenge for executives as they carefully allocate marginal dollars toward value creation.

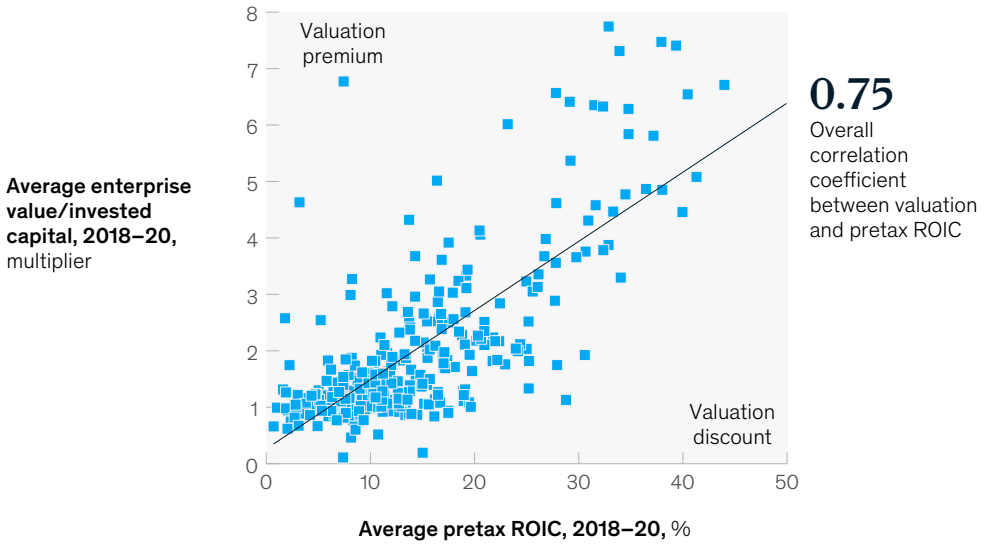
Companies can improve their odds of success by focusing on areas of the project life cycle—capital strategy and portfolio optimization, project development and value improvement, and project delivery and construction—while investing in foundational enablers.

Cracking the code on capital expenditure management

Despite the importance of capital expenditure management in executing business strategy, preserving cash, and maximizing ROIC, most companies struggle in this area for two primary

Exhibit 1

The strong correlation between pretax ROIC and valuation highlights the importance of wise capital expenditure strategy.



Source: Company filings, S&P Capital IQ

reasons. First, capital expenditure is often not a core business; instead, organizations focus on operating performance, where they have extensive institutional knowledge. When it comes to capital projects, executives rely on a select few people with experience in capital delivery. Second, capital performance is typically a black box. Executives find it difficult to understand and predict the performance of individual projects and the capital project portfolio as a whole.

Across industries, we see companies struggle to deliver projects on time and on schedule (Exhibit 2). In fact, cost and schedule overruns compared with original estimates frequently exceed 50 percent. Notably, these occur in both the public and private sectors.

The COVID-19 pandemic has accelerated and magnified these challenges. Governments are increasingly viewing infrastructure spending as a tool for economic stimulus, which amplifies the cyclical nature of capital expenditure deployments. At the same time, some organizations have had to make drastic cutbacks in capital projects because of difficult economic conditions. The reliance on just a few experienced people when travel restrictions necessitated a remote-operating model further increased the complexity. As a result, only a few organizations have been able to maintain a through-cycle perspective.

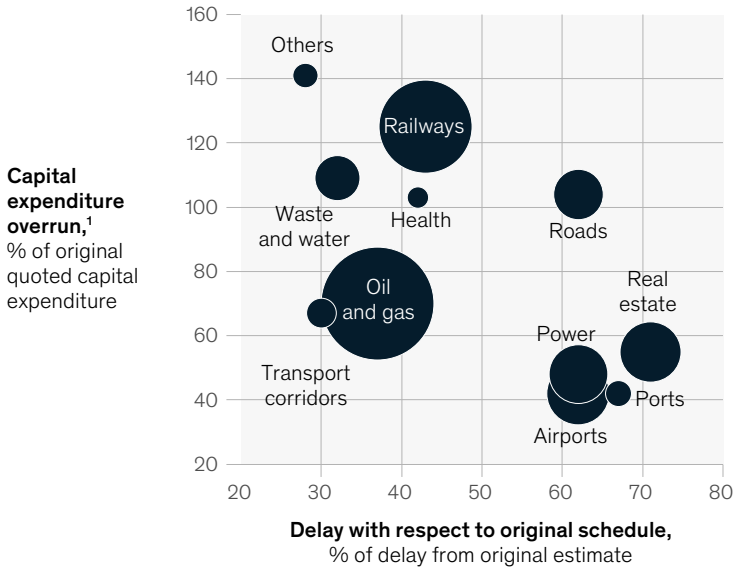
In addition, current inflation could put an end to the historically low interest rates that companies are enjoying for financing their projects. As the cost of capital goes up, discipline in managing large projects will become increasingly important.

Improving capital expenditure management

In our experience, the organizational drivers that impede capital expenditure management affect all stages of a project life cycle, from

Exhibit 2

Most capital projects experience significant time and cost overruns.



$1.21 billion
Average cost overrun of projects[2]

79%
Cost overruns on average relative to initial budget[2]

52%
Schedule delays on average compared with initial schedule[1]

5.4%
Projects that meet both authorized cost and schedule, according to the Construction Industry Institute

[1] Based on a sample of 427 projects.
[2] Based on a sample of 532 projects.
Source: McKinsey analysis

portfolio management to project execution and commissioning. Best-in-class capital development and delivery require companies to outperform in three main areas, supported by several foundational enablers (Exhibit 3).

Recipes for capturing value

Companies can transform the life cycle of a capital expenditure project by focusing on three areas: capital strategy and portfolio optimization, project development and value improvement, and project delivery and construction. While the savings potential applies to each area on a stand-alone basis, their impact has some overlap. In our experience, companies that deploy these best practices are able to save 15 to 30 percent of a project's cost.

Capital strategy and portfolio optimization

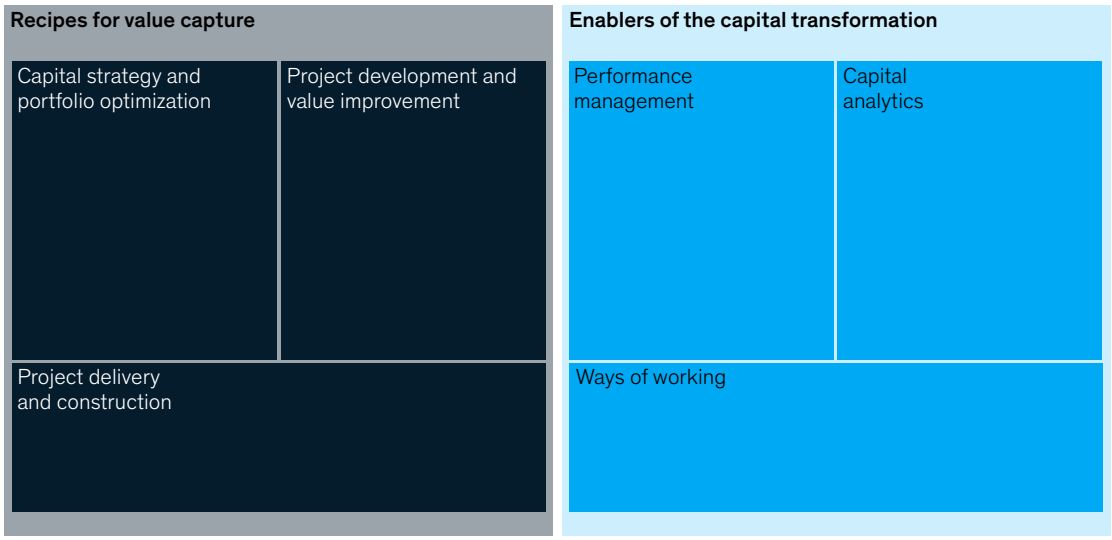
The greatest opportunity to influence a project's outcome comes at its start. Too often, organizations commit to projects without a proper understanding of business needs, incurring significant expense to deliver an outcome misaligned with the overall strategy. Indeed, a failure to adequately recognize, price, and manage the inherent risks of project delivery is a recurring issue in the industry. Organizations can address this challenge by following a systematic three-step approach:

Assess the current state of capital projects and portfolio. It's essential to identify strengths, areas of improvement, and the value at stake. To do so, organizations must build a transparent and rigorously tested baseline and capital budget, which should provide a clear understanding of the overall capital expenditure budget for the coming years as well as accurate cost and time forecasts for an organization's portfolio of capital projects.

Ensure capital allocation is linked to overall company strategy. This step involves reviewing sources and uses of cash and ensuring allocated capital is linked to strategy. Companies must set an enterprise-wide strategy,[1] assess the current portfolio against the relevant market with forward-looking assessments and cash flow simulation, and review sources and uses of cash to determine the

Exhibit 3

A structured approach to capital expenditure management includes recipes for value capture and enablers.



[1] For more, see "Strategy to Beat the Odds," McKinsey.

amount of capital available. Particular focus should be given to environmental, social, and governance (ESG) considerations—by both proactively managing risks and capturing the full upside opportunity of new projects—because sustainability is becoming a real source of shareholder value (Exhibit 4). With this knowledge, organizations can identify internal and external opportunities to strengthen their portfolio based on affordability and strategic objectives.

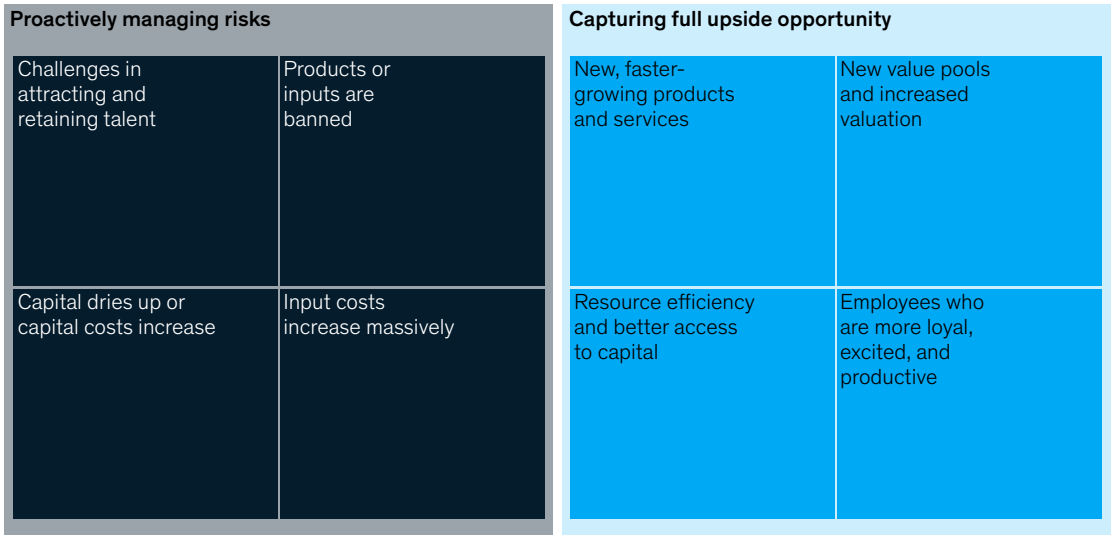
Optimize the capital portfolio to increase company-wide ROIC. Executives should distinguish between projects that are existing or committed, planned and necessary (for legal, regulatory, or strategic requirements), and discretionary. They can do so by challenging a project's justification, classifications, benefit estimates, and assumptions to ensure they are realistic. This analysis helps companies to define and calibrate their portfolios by prioritizing projects based on KPIs and discussing critical projects not in the portfolio. Executives can then verify that the portfolio is aligned with the business strategy, risk profile, and funding constraints.

For example, a commercial vehicle manufacturer recently undertook a rigorous review of its project portfolio. After establishing a detailed baseline covering several hundred planned projects in one data set, the manufacturer classified the projects into two categories: must-have and discretionary. It also considered strategic realignment in light of a shift to e-mobility and the implications on investments in internal-combustion-engine vehicles. Last, it scrutinized individual maintenance projects to reduce their scope. Overall, the manufacturer uncovered opportunities to decrease its capital expenditure budget by as much as 20 percent. This strict review process became part of its annual routine.

Project development and value improvement
While value-engineering exercises are common, we find that 5 to 15 percent of additional value is typically left on the table. Too often, organizations

Exhibit 4

Sustainability is becoming a real source of shareholder value through the lowering of risks and capturing of value.



30–50%
Corporate profits at stake from external engagement; eg, carbon tax

6×
Growth in sustainability products; eg, circular plastics and low-carbon products

focus on technical systems and incremental improvements. Instead, executives should consider the full life cycle cost across several areas:

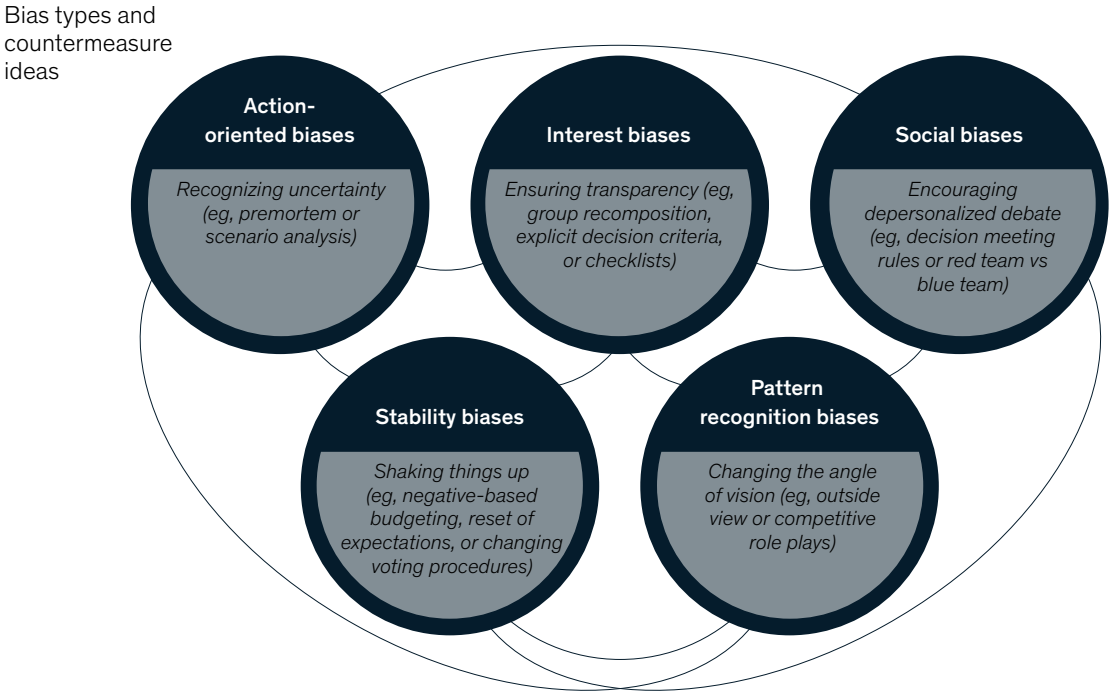
Sourcing the right projects with the right partners. Companies must ensure they are sourcing the right projects by aligning on prioritization criteria and identifying the sectors to play in based on their strategy. Once these selections are made, organizations can use benchmarking and advanced-analytics tools to accelerate project timelines and improve planning. Building the right consortium of contractors and partners at the outset and establishing governance and reporting can have a huge impact. Best-in-class teams secure the optimal financing, which can include public and private sources, by assessing the economic, legal, and operational implications for each option.

A critical success factor is a strong tendering office, which focuses on choosing better projects. It can increase the likelihood of winning through better partnerships and customer insights and enhance the profitability of bids with creative solutions for reducing cost and risk. Best-in-class tendering offices identify projects aligned with the company's strategy, have a clear understanding of success factors, develop effective partnerships across the value chain, and implement a risk-adjusted approach to pricing.

Achieve the full potential of the preconstruction project value. Companies can take a range of actions to strengthen capital effectiveness. For example, they should consider the project holistically, including technical systems, management systems, and mindsets and behaviors. To ensure they create value across all stages of the project life cycle, organizations should design contract and procurement interventions early in the project. An emphasis on existing ideas and proven solutions can help companies avoid getting

Exhibit 5

World-class organizations formalize dedicated systems and processes to reduce biased decision making and combat five types of biases with proven techniques.



bogged down in developing new solutions. For instance, a minimum-technical-solution approach can be used to identify the highest-value projects by challenging technical requirements once macro-elements are confirmed.

Companies should also seek to formalize dedicated systems and processes to support decision making and combat bias. We have identified five types of biases to which organizations should pay close attention (Exhibit 5). For instance, interest biases should be addressed by increasing transparency in decision making and aligning on explicit decision criteria before assessing the project. Stability biases can also be harmful. We have seen it too many times: companies have a number of underperforming projects that just won't die and that take up valuable and already limited available resources. Organizations should invest in quickly determining when to halt projects—and actually stop them.

Setting up a system to take action in a nonbiased way is a crucial element of best-in-class portfolio optimization. Changing the burden of proof can also help. One energy company counterbalanced the natural desire of executives to hang on to underperforming assets with a systematic process for continually upgrading the company's portfolio. Every year, the CEO asked the corporate-planning team to identify 3 to 5 percent of the company's assets that could be divested. The divisions could retain any assets placed in this group but only if they could demonstrate a compelling turnaround program for them. The burden of proof was on the business units to prove that an asset should be retained, rather than just assuming it should.

An effective governance system ensures that all ideas generated from project value improvements are subject to robust tracking and follow-up. Further, the adoption of innovative digital and technological solutions can enhance standardization, modularization, transparency, and efficiency. A power company recently explored options to phase out coal-powered energy using a project value improvement methodology and a minimum technical solution. The process helped to articulate options to maximize ROI and minimize greenhouse-gas emissions. An analysis of each option, using an idea bank of more than 2,000 detailed ideas, let the company find solutions to reduce investment on features with little value added, reallocate spending to more efficient technologies, and better adjust capacity configurations with business needs. Ultimately, the company reduced capital costs by 30 percent while increasing CO_2 abatement by the same amount.

Designing the right project organization. An open, collaborative, and result-focused environment enabled by stringent performance management processes is critical for success, regardless of the contractual arrangement between owners and contractors. Improving capital project practices is possible only if companies measure those practices and understand where they stand compared with their peers. The organization should be designed with a five-year capital portfolio in mind and built by developing structures for project archetypes and modeling the resources required to deliver the capital plan. A rigorous stage-gate process of formal reviews should also be implemented to verify the quality of projects moving forward. Too many projects are rushed through phases with no formal review of their deliverables, leading to a highly risky execution phase, which usually results in delays and cost overruns.

As successful organizations demonstrate, addressing organizational health in project teams is as important as performance initiatives. McKinsey research has found that the healthiest organizations generate three times higher returns than companies in the bottom quartile and more than 60 percent higher returns compared with companies in the middle two quartiles.[2]

Project delivery and construction

Since the root causes of poor performance—project complexity, data quality, execution capabilities, and incentives and mindsets—can be difficult to identify and act on, organizations can benefit from taking the following actions across project delivery and construction dimensions.

[2] Scott Keller and Bill Schaninger, *Beyond Performance 2.0: A Proven Approach to Leading Large-Scale Change*, second edition, Hoboken, NJ: Wiley, 2019.

Optimize the project execution plan. Organizations should embrace principles of operations science to develop an optimized configuration for the production system, as well as set a competitive and realistic baseline for the project. This execution plan identifies the execution options that could be deployed on the project and key decisions that need to be made. Companies should also break the execution plan into its microproduction systems and visualize the complicated schedule. Approaching capital projects as systems allows companies to apply operations science across process design, capacity, inventory, and variability.

Contract, claims, and change orders management. While claims are quite common on capital projects, proactive management can keep them under control and allow owners to retain significant value. Focusing on claims avoidance when drafting terms and conditions can head off many claims before they arise. In addition, partnering with contractors creates a more collaborative environment, making them less inclined to pursue an aggressive claims strategy. To manage change orders on a project, companies should address their contract management capability, project execution change management, and project closeout negotiation support. A European chemical company planning to build greenfield infrastructure in a new Asian geography recently employed this approach. It reduced risk on the project by bringing together bottom-up, integrated planning and performance management with targeted lean-construction interventions. By doing so, the company reduced the project's duration by a year, achieved on-time delivery, and stayed within its €1 billion budget.

Enablers of the capital transformation

These three value capture areas must be supported by a capable organization with the right tools and processes—what we call the "transformational chassis." To establish this infrastructure, organizations should focus on several activities.

Performance management

The best organizations institute a performance management system to implement a cascading set of project review meetings focused on assessing the progress of value-creation initiatives. Building on a foundation of quality data, the right performance conversations must take place at all levels of the organization.

Companies should also be prepared to reexamine their stage-gate governance system to shift from an assurance mindset (often drowning in bureaucracy and needless reporting) to an investor mindset. Critical value-enabling activities should be defined at each stage of the project life cycle, supported by a playbook of best practices for execution and implemented by a project review board. While governance processes exist, they often involve reporting without decision making or are not focused on the right outcomes—for example, ensuring that the investment decision and thesis remain valid through a project's life. Quite often, companies provide incentives for project managers to execute an outdated project plan rather than deliver against the organization's needs and goals.

Creating project transparency is also critical. Companies should establish a digital nerve center—or control tower—that collects field-level data to establish a single source of truth and implement predictive analytics. Equally important, companies must address capability building to ensure that the team has a solid understanding of the baseline and embraces data-based decision making.

Companies should stand up delivery teams that integrate owner and contractor groups across disciplines and institute a consistent and effective project management rhythm that can identify risks and opportunities over a project's duration. Once delivery teams prioritize the biggest opportunities, dedicated capacity should be allocated to solve a project's most challenging problems. Finally, companies should build and deploy comprehensive programs that improve culture and workforce

capabilities throughout the organization, including the front line.

Capital analytics

Many organizations struggle to get a clear view of how projects are performing, which limits the possibility for timely interventions, decision making, and resource planning. By digitalizing the performance management of construction projects using timely and transparent project data, companies can track value capture and leading indicators while making data available across the enterprise. Using a single source of truth can reduce delivery risk, increase responsiveness, and enable a more proactive approach to the identification of issues and the capture of opportunities. The most advanced projects build automated, real-time control towers that consolidate information across systems, engineering disciplines, project sites, contractors, and broader stakeholders. The ability to integrate data sets speeds decision making, unlocks further insights, and promotes collaborative problem solving between the company that owns the capital project and the engineering, procurement, and construction company.

Ways of working

In many cases, executives are unwilling to engage in comprehensive capital reviews because they lack a sufficient understanding of capital management processes, and project managers can be afraid to expose this lack of proficiency. Agile practices can facilitate rapid and effective decision making by bringing together cross-functional project teams. Under this approach, organizations establish daily stand-ups, weekly showcases, and fortnightly sprints to help eliminate silos and maintain a focus on top priorities. Agility must be supported by an organizational structure, well-developed team capabilities, and an investment mindset. Organizations should also build skills and establish a culture of cooperation to optimize their capital investments.

We do recognize that getting capital expenditure management right feels like a lot to do well. And although many of these tasks are somehow done by a slew of companies, pockets of organizational excellence can be undermined instantly (and sometimes existentially) by one big project that goes wrong or a strategic misfire that pushes an organization from being a leader to a laggard in the investment cycle. In some ways, capital expenditure management leaders face similar challenges to those in other functions that have already undergone major productivity improvements: often these challenges are not technical problems but instead relate to how people work together toward a common goal.

Yet we believe organizations have a significant opportunity to fundamentally improve project outcomes by rethinking traditional approaches to project delivery. Sustainable improvements can be achieved by resizing the project portfolio, optimizing the cash flows for individual projects, and improving and reducing individual project delivery risk.

Tom Brinded is a partner in McKinsey's London office, **Erikhans Kok** is a partner in the Houston office, **Lucas Ponbauer** is a partner in the Zurich office, and **Bevan Watson** is a partner in the Sydney office.

The authors wish to thank Michael Gootman, Nikolaus Lehmann, and Andrés Saint-Jean for their contributions to this article.

Designed by McKinsey Global Publishing
Copyright © 2022 McKinsey & Company. All rights reserved.